

Aditya Kulkarni

adityakulkarnius@gmail.com | (281) 876-7066 | linkedin.com/in/aditya-kulkarni-355b81217 | github.com/Aditya-k24 | kulkarniaditya.com | Raleigh, NC

SUMMARY

Applied ML engineer with hands-on experience building LLM evaluation infrastructure, data pipelines for model workflows, and production AI systems. Built a rigorous vendor-neutral eval harness spanning 2,000 frozen examples with statistical significance testing -- the kind of infrastructure Reflection needs to measure whether fine-tuned models genuinely improve. Complementary background in distributed data pipelines (Hadoop, Spark, Kafka) and reproducible ML research.

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Built multi-model LLM routing over 8+ models (Claude, GPT-4o, Gemini) via MCP; implemented Temporal.io durable orchestration for document processing pipelines; 394 commits, 7 production deployments in TypeScript/Node.js/PostgreSQL/Redis.
 - Designed and shipped real-time AI features including Auth0 RBAC, Pusher event push, and Sentry observability across 10+ backend features.
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Built cloud-native genomic ML platform on Google Cloud Run (500 GB dataset, 100+ researchers); improved model pipeline accuracy 12% via systematic hyperparameter tuning and reproducible experiment tracking.
- Software Developer Intern

Mphasis (HP)

Jun - Aug 2024

 - Delivered health-claim ML classification tool (+25% accuracy via OCR/NLP); automated AWS/GCP CI/CD and added Grafana monitoring.
- AI/ML Intern

Capgemini

Jun - Aug 2023

 - Built Hadoop+Spark batch data pipelines and Kafka real-time ingestion for ML model training data; reduced processing time 15% and latency 20%.

PROJECTS

- LLM Eval Harness

Evaluation Infrastructure

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, Gemini 2.5 Pro on 2,000 frozen examples (grounded QA, multi-hop QA, FEVER claim verification). Tracked hallucination rate, TokenF1, p50/p95 latency, cost/call; paired bootstrap CIs, McNemar test, Holm correction to guard against benchmark overfitting. DuckDB + Parquet + Streamlit dashboard.

Python · DuckDB · Parquet · Streamlit · PyTorch · Hugging Face
- Glance AI

RAG / Data Pipelines

Compliance-ready RAG system for fintech/healthcare; per-tenant data isolation at vector retrieval layer; deterministic output architecture; reproducible data preparation pipeline with quality validation.

Python · LangChain · RAG · FastAPI · PostgreSQL · pgvector
- VectorLift

Neural IR / ML Serving

BM25 + dense bi-encoder + cross-encoder re-ranker on MS MARCO; Kafka-driven indexing into Elasticsearch + Qdrant; FastAPI serving; NDCG@10/MRR@10 evaluation; Prometheus + Grafana observability.

Python · Elasticsearch · Qdrant · Kafka · FastAPI · Prometheus
- Mapify

IEEE Published

T5-transformer pipeline converting 1,500+ PDFs/images to structured mind maps; reduced manual review time 60%; presented to 500+ attendees at IEEE conference.

Python · T5 · Hugging Face · OCR · NLP

EDUCATION

- MS Computer Science — NC State University

Aug 2024 - May 2026

AI, Data Science, Software Engineering, Algorithms
- BTech Computer Science — Mumbai University

Aug 2020 - May 2024

CGPA 9.32 / 10

SKILLS & AWARDS

- Languages:

Python, TypeScript, Java, SQL

ML/AI:

LLMs, RAG, LLM Evals, PyTorch, Hugging Face, Transformers, NLP, Embeddings
- Data/Infra:

Hadoop, Spark, Kafka, PostgreSQL, Redis, Docker, AWS, GCP, Prometheus, Grafana, GitHub Actions
- HackNC24 Top 10 / 100+ teams

IEEE Published (Mapify, 500+ attendees)

Sanjeevani Top 1% / 1,100+ submissions

Google DSC Lead (500+ students)